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- Roll No : SY B 23
- Batch : SY B2

Aim

To perform data wrangling operations using Pandas such as filtering data, grouping, merging datasets, handling missing values, and applying transformations to prepare data for analysis.

Theory

1. What is Data Wrangling?

Data wrangling is the process of cleaning, organizing, and transforming raw data into a useful format for analysis. It includes handling missing values, correcting errors, and restructuring data. Data preprocessing is a part of data wrangling that mainly focuses on preparing data for machine learning models, while data wrangling is a broader process that includes cleaning and transforming data for general use.

2. Importance of EDA in Pandas

Exploratory Data Analysis (EDA) helps us understand the dataset before applying any model. Using Pandas, we can easily check data structure, detect missing values, find patterns, and summarize data. It helps in making better decisions and improving data quality.

3. Difference between Series, DataFrame, and Panel

A Series is a one-dimensional array that holds data like a single column. A DataFrame is a two-dimensional table with rows and columns. A Panel is a three-dimensional data structure, but it is now removed in newer versions of Pandas and replaced by other methods.

4. Functions in Pandas

- a. `isnull()` It is used to check missing values in the dataset and returns True if the value is missing.
- b. `notnull()` It returns True for non-missing values and False for missing values.
- c. `dropna()` It removes rows or columns that contain missing values.

d. fillna() It fills missing values with a specified value like mean, median, or any constant.

5. Column Transformation

Column transformation means modifying the values of a column using operations or functions. It is important because it helps in converting data into a suitable format for analysis, like scaling values or creating new features.

```
In [ ]: import pandas as pd
df = pd.read_csv("/content/employee updated (1).csv")
print(df)

# 1. Select a single column from a DataFrame
name = df['Name']
print(name.head(5))

# 2. Add a new column with constant values
df['Constant_Value'] = 10
print(df.head())

# 3. Sort a DataFrame by one column
df_sorted_age = df.sort_values('Age')
print(df_sorted_age.head(5))

# 4. Rename DataFrame columns
df_renamed = df.rename(columns={'Employee_ID': 'ID'})
print(df_renamed.head())

# 5. Display summary statistics using describe()
print(df.describe())

# 6. Remove duplicate rows from a dataset
df_no_duplicates = df.drop_duplicates()
print(df_no_duplicates.head(10))
display(df_renamed.head())

# 7. Handle missing values using mean or median
print("Handling missing data")
df.loc[2, 'Salary'] = None # Introducing NaN (missing value) in the 'Score' c
print(df)
df['Salary'] = df['Salary'].fillna(df['Salary'].mean())
print("\nAfter filling missing 'Score' values with mean:")
print(df)

# First, let's re-introduce a missing value in 'Bonus' for demonstration
df.loc[0, 'Bonus'] = None

print("Missing values before handling with mean and median:")
print(df[['Salary', 'Bonus']].isnull().sum())

# Fill missing 'Salary' values with the mean
```

```

df['Salary'] = df['Salary'].fillna(df['Salary'].mean())

# Fill missing 'Bonus' values with the median
df['Bonus'] = df['Bonus'].fillna(df['Bonus'].median())

print("\nMissing values after handling with mean and median:")
print(df[['Salary', 'Bonus']].isnull().sum())

print(df.head())

# 8. Change data types of selected columns
print("Data types before change:\n", df.dtypes)
# Convert 'Join_Date' to datetime type
df['Join_Date'] = pd.to_datetime(df['Join_Date'])
# Convert 'Gender', 'Department', and 'Experience/Fresher' to categorical type
if 'Gender' in df.columns:
    df['Gender'] = df['Gender'].astype('category')
if 'Department' in df.columns:
    df['Department'] = df['Department'].astype('category')
if 'Experience/Fresher' in df.columns:
    df['Experience/Fresher'] = df['Experience/Fresher'].astype('category')

print("\nData types after change:\n", df.dtypes)
print(df.head())

# 9. Filter rows based on a given condition
# Filter employees older than 30 years
df_filtered_age = df[df['Age'] > 30]
print(f"Number of employees older than 30: {len(df_filtered_age)}")
print(df_filtered_age.head())

# 10. Save the cleaned dataset to a new CSV file
df.to_csv('employee_cleaned.csv', index=False)
print("Data saved to 'employee_cleaned.csv'")

```

	Employee_ID	Name	Age	Gender	Department	Join_Date	Salary \
0	101	Alice	50	Female	Marketing	2009-01-18	62606.000000
1	102	Bob	36	Female	Marketing	2012-01-01	41534.000000
2	103	Charlie	29	Female	IT	2007-02-15	76338.368421
3	104	David	42	Male	Marketing	2015-12-29	31016.000000
4	105	Eve	40	Female	IT	2005-02-04	119789.000000
5	106	Frank	44	Male	Marketing	2013-08-19	85591.000000
6	107	Grace	32	Male	Marketing	2010-05-10	119812.000000
7	108	Hank	32	Male	IT	2009-05-05	53247.000000
8	109	Ivy	45	Male	IT	2015-10-19	54300.000000
9	110	Jack	57	Male	Marketing	2013-06-01	104065.000000
10	111	Kate	45	Female	HR	2007-10-19	112798.000000
11	112	Leo	24	Female	IT	2014-06-24	39268.000000
12	113	Mona	43	Female	HR	2012-02-27	116807.000000
13	114	Nina	23	Female	HR	2019-04-23	42185.000000
14	115	Oscar	45	Female	HR	2015-07-11	93704.000000
15	116	Paul	51	Male	IT	2009-02-09	116779.000000
16	117	Quinn	59	Female	HR	2018-02-20	69099.000000
17	118	Rose	23	Female	IT	2005-06-11	38571.000000
18	119	Steve	42	Male	HR	2016-10-07	68044.000000
19	120	Tina	54	Female	Marketing	2010-06-05	81214.000000

	Experience	Performance_Rating	Bonus	Work_Hours_Per_Week \
0	13.000000	2.000000	9488.000000	44.0
1	1.000000	1.000000	5206.000000	40.0
2	6.388889	2.722222	11083.388889	39.5
3	7.000000	1.000000	10977.000000	42.0
4	9.000000	4.000000	12721.000000	36.0
5	6.388889	2.000000	12035.000000	48.0
6	1.000000	1.000000	6484.000000	31.0
7	12.000000	5.000000	12858.000000	39.0
8	8.000000	2.722222	5863.000000	42.0
9	8.000000	4.000000	7790.000000	35.0
10	11.000000	3.000000	12408.000000	41.0
11	3.000000	3.000000	13755.000000	41.0
12	1.000000	1.000000	10116.000000	49.0
13	8.000000	3.000000	17383.000000	40.0
14	3.000000	5.000000	11083.388889	36.0
15	3.000000	3.000000	6757.000000	30.0
16	1.000000	1.000000	19820.000000	30.0
17	11.000000	5.000000	12574.000000	49.0
18	5.000000	2.000000	11374.000000	39.5
19	10.000000	3.000000	11892.000000	38.0

	Experience/Fresher
0	Experienced
1	Fresher
2	Fresher
3	Experienced
4	Experienced
5	Experienced
6	Fresher
7	Fresher
8	Experienced

9 Experienced
 10 Experienced
 11 Fresher
 12 Experienced
 13 Fresher
 14 Experienced
 15 Experienced
 16 Experienced
 17 Fresher
 18 Experienced
 19 Experienced

0 Alice
 1 Bob
 2 Charlie
 3 David
 4 Eve

Name: Name, dtype: object

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	Experience/Fresher	Constant_Value
0	Experienced	10
1	Fresher	10
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3	Experienced	10
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13	114	Nina	23	Female	HR	2019-04-23	42185.000000
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2	103	Charlie	29	Female	IT	2007-02-15	76338.368421
6	107	Grace	32	Male	Marketing	2010-05-10	119812.000000

	Experience	Performance_Rating	Bonus	Work_Hours_Per_Week \
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6	1.000000	1.000000	6484.000000	31.0

	Experience/Fresher	Constant_Value
13	Fresher	10
17	Fresher	10

11		Fresher		10			
2		Fresher		10			
6		Fresher		10			
	ID	Name	Age	Gender	Department	Join_Date	Salary \
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	Experience/Fresher	Constant_Value
0	Experienced	10
1	Fresher	10
2	Fresher	10
3	Experienced	10
4	Experienced	10

	Employee_ID	Age	Salary	Experience	Performance_Rating \
count	20.000000	20.000000	20.000000	20.000000	20.000000
mean	110.500000	40.800000	76338.368421	6.388889	2.722222
std	5.91608	10.909195	30882.861701	3.962175	1.369039
min	101.000000	23.000000	31016.000000	1.000000	1.000000
25%	105.750000	32.000000	50481.500000	3.000000	1.750000
50%	110.500000	42.500000	72718.684211	6.694444	2.861111
75%	115.250000	46.250000	106248.250000	9.250000	3.250000
max	120.000000	59.000000	119812.000000	13.000000	5.000000

	Bonus	Work_Hours_Per_Week	Constant_Value
count	20.000000	20.000000	20.0
mean	11083.388889	39.500000	10.0
std	3631.799585	5.57485	0.0
min	5206.000000	30.000000	10.0
25%	9063.500000	36.000000	10.0
50%	11228.694444	39.750000	10.0
75%	12610.750000	42.000000	10.0
max	19820.000000	49.000000	10.0

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5	106	Frank	44	Male	Marketing	2013-08-19	85591.000000
6	107	Grace	32	Male	Marketing	2010-05-10	119812.000000
7	108	Hank	32	Male	IT	2009-05-05	53247.000000
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2	103	Charlie	29	Female	IT	2007-02-15	76338.368421	6.388889
3	104	David	42	Male	Marketing	2015-12-29	31016.000000	7.000000
4	105	Eve	40	Female	IT	2005-02-04	119789.000000	9.000000

Handling missing data

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1	102	Bob	36	Female	Marketing	2012-01-01	41534.0
2	103	Charlie	29	Female	IT	2007-02-15	NaN
3	104	David	42	Male	Marketing	2015-12-29	31016.0
4	105	Eve	40	Female	IT	2005-02-04	119789.0
5	106	Frank	44	Male	Marketing	2013-08-19	85591.0
6	107	Grace	32	Male	Marketing	2010-05-10	119812.0
7	108	Hank	32	Male	IT	2009-05-05	53247.0
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10	111	Kate	45	Female	HR	2007-10-19	112798.0
11	112	Leo	24	Female	IT	2014-06-24	39268.0
12	113	Mona	43	Female	HR	2012-02-27	116807.0
13	114	Nina	23	Female	HR	2019-04-23	42185.0
14	115	Oscar	45	Female	HR	2015-07-11	93704.0
15	116	Paul	51	Male	IT	2009-02-09	116779.0
16	117	Quinn	59	Female	HR	2018-02-20	69099.0
17	118	Rose	23	Female	IT	2005-06-11	38571.0
18	119	Steve	42	Male	HR	2016-10-07	68044.0
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	Experience/Fresher	Constant_Value
0	Experienced	10
1	Fresher	10
2	Fresher	10
3	Experienced	10
4	Experienced	10
5	Experienced	10
6	Fresher	10
7	Fresher	10

8	Experienced	10
9	Experienced	10
10	Experienced	10
11	Fresher	10
12	Experienced	10
13	Fresher	10
14	Experienced	10
15	Experienced	10
16	Experienced	10
17	Fresher	10
18	Experienced	10
19	Experienced	10

After filling missing 'Score' values with mean:

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17	11.000000	5.000000	12574.000000	49.0
18	5.000000	2.000000	11374.000000	39.5
19	10.000000	3.000000	11892.000000	38.0

	Experience/Fresher	Constant_Value
0	Experienced	10
1	Fresher	10
2	Fresher	10
3	Experienced	10
4	Experienced	10
5	Experienced	10
6	Fresher	10
7	Fresher	10
8	Experienced	10
9	Experienced	10
10	Experienced	10
11	Fresher	10
12	Experienced	10
13	Fresher	10
14	Experienced	10
15	Experienced	10
16	Experienced	10
17	Fresher	10
18	Experienced	10
19	Experienced	10

Missing values before handling with mean and median:

Salary 0
 Bonus 1
 dtype: int64

Missing values after handling with mean and median:

Salary 0
 Bonus 0
 dtype: int64

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	Experience/Fresher	Constant_Value
0	Experienced	10
1	Fresher	10
2	Fresher	10
3	Experienced	10
4	Experienced	10

Data types before change:

```
Employee_ID      int64
Name              object
Age              int64
Gender            object
Department        object
Join_Date         object
Salary            float64
Experience         float64
Performance_Rating float64
Bonus             float64
Work_Hours_Per_Week float64
Experience/Fresher object
Constant_Value    int64
dtype: object
```

Data types after change:

```
Employee_ID      int64
Name              object
Age              int64
Gender            category
Department        category
Join_Date         datetime64[ns]
Salary            float64
Experience         float64
Performance_Rating float64
Bonus             float64
Work_Hours_Per_Week float64
Experience/Fresher category
Constant_Value    int64
dtype: object
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2	Fresher	10
3	Experienced	10
4	Experienced	10

Number of employees older than 30: 16

	Employee_ID	Name	Age	Gender	Department	Join_Date	Salary \
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	Experience/Fresher	Constant_Value
0	Experienced	10
1	Fresher	10
3	Experienced	10
4	Experienced	10
5	Experienced	10

Data saved to 'employees cleaned .csv'

Conclusion

In this assignment, we performed different data wrangling operations using Pandas in a simple and practical way. We learned how to filter data using multiple conditions, group data to calculate average values, and merge different datasets based on a common column. We also understood how to remove unnecessary rows and columns and apply functions to modify data. These operations help in cleaning and organizing raw data, which is very important for accurate data analysis. Overall, data wrangling makes the dataset more useful, structured, and ready for further analysis.